Sr.No. 8519

AY: 2025-26
Session: Odd MSE October 2025
Date: 08-10-2025
Sem: 1
Slot: 10:00am - 11:00am
C. Code: CS413
Exam Type: Written
Block:4D1
Course Section:1
Seat No: 2022600044
1_29_501_986_1_1505_38346

Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHONOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058

MID SEMESTER EXAMINATION

Name of the candidate. ADITI PILLAI

Candidate's UID number: 2 0 2 2 6 0 0 0 4 4

Examination class room number: 4 0 1

Invigilator's signature with date: 8/10/25

(To be filled in by the candidate)

EXAMINATION: BE MSE
(For example FY MCA/FY BTECH)

PROGRAM: CSE-AIML

DAY AND DATE: WEDNESDAY, 08/10/2025

COURSE NAME: DEEP LEARNING (PE)

SEMESTER :

I / II / III / IV / V / VI / VII / VIII

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

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INSTRUCTIONS TO CANDIDATES

1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.

2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.

3) Do not write anything in the column provided for the marks to be assigned by the examiners.

4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.

5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.

6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.

7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.

8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**

9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.

10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.

11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.

12) The answer-books will be scrutinized before they are sent to examiners.

13) All answer-books supplied shall be returned whether written or blank.

14) Nothing shall be written on the question-paper.

15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.

16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.

17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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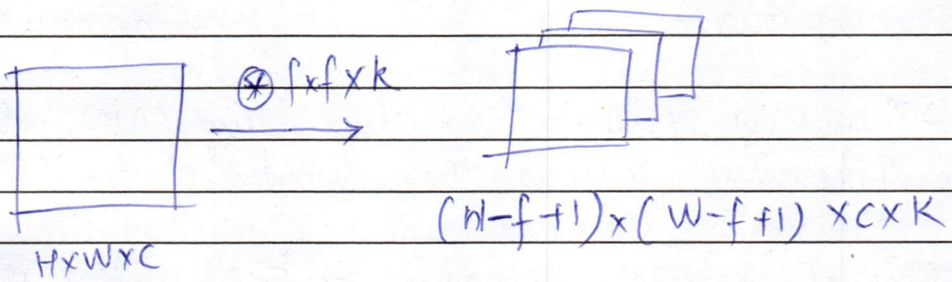
~~Q1 a) $f(x) = \frac{1}{2}\|Xw - y\|^2$~~

Q2] a] 2D convolution

Input : $H \times W \times C$
Filters : K filters of size $F \times F$

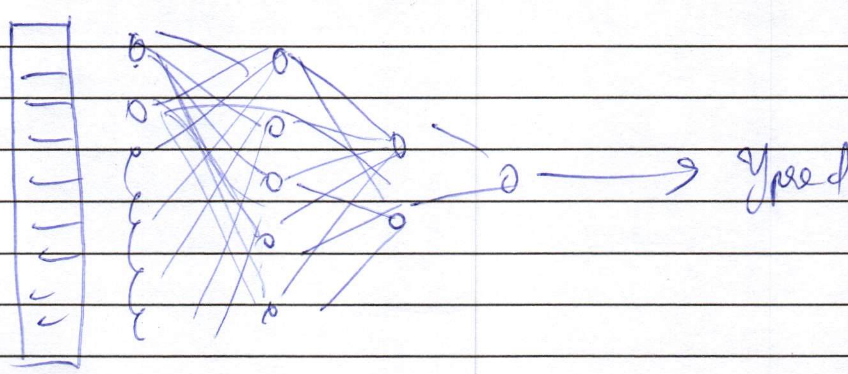
$(n-f+1)$.

Convolution Architecture:-



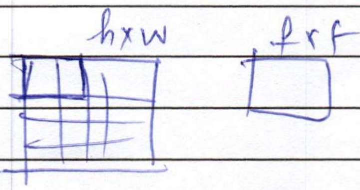
c images given as i/p.

this will be flattened into a 1d input and passed to the fully connected layers.



$(H-f+1)(W-f+1)ck \times 1$

∴ No. of multiplications :



$= \left(\frac{H}{f} \times \frac{W}{f}\right) \times C \times \underbrace{K}_{\text{no. of images}}$ which leads to filtered images of size $(H-f+1) \times (W-f+1) \times \underbrace{C \times k}_{\text{no. of images}}$.

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→ Depthwise separable convolutions are critical in modern architectures because they have lesser parameters, implying that they can be trained in lesser time.

→ In ANN, the weights and bias tend to increase the computation time ~~as~~ due to more trainable parameters. Thus, if we try to achieve the same result~~s~~ on images which have a ~~cess~~ huge amount of pixels, the computation lag will be very apparent.

→ Thus, we rectify this by using CNN, which efficiently reduces the number of trainable parameters by using filters.

→ These filters are important as they help detect edges, shapes. Each filter performs a specific task, and together, CNN can make accurate predictions.

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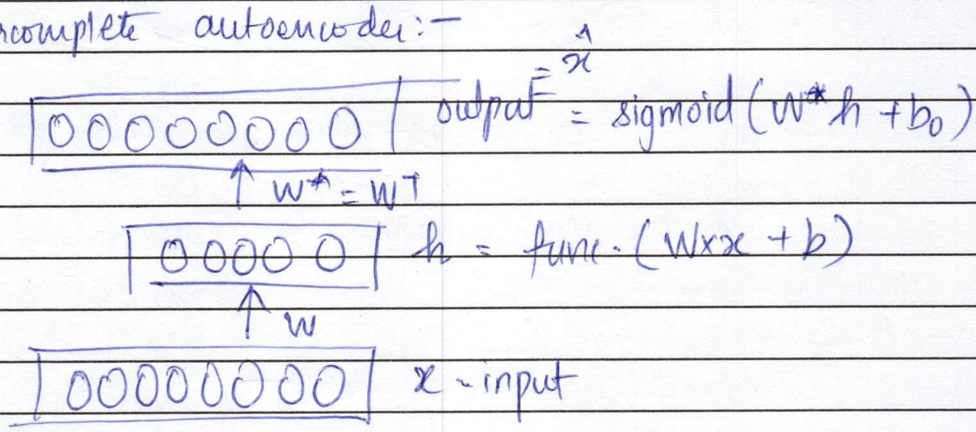
Q2]b] Undercomplete autoencoder:-



Linear $\Rightarrow$ $h = (Wx + b)$ ~~MSE~~ $\sum(x$

The undercomplete autoencoder is where the hidden layer has less neurons than the input layer. Thus, The hidden layer has to recognize the important features of the input x to encode, implying that it has to choose what input to take into consideration. Thus, this is equivalent to a PCA.

Linear: $\Rightarrow$ $h = Wx + b_1$

$\cancel{\hat{x}}\ \hat{x} = W^T x + b_2$

$$MSE = \frac{1}{n} \sum (\hat{x}_i - x_i)^2$$

$$= \frac{1}{n} \sum_{i=1}^{n} (W^T x + b_2 - Wx + b_1)$$

We want to minimize the error.

$\Rightarrow$ derivative of $\frac{1}{n} \sum (W^T - W)x + \cancel{\frac{n(b_2 - b_1)}{x}} = 0$

$$\therefore \frac{\partial}{\partial x} \sum (W^T - W)x = 0$$

$$(W^T - W) = 0$$

$$\therefore (W^T W^{-1} - 1) = I$$

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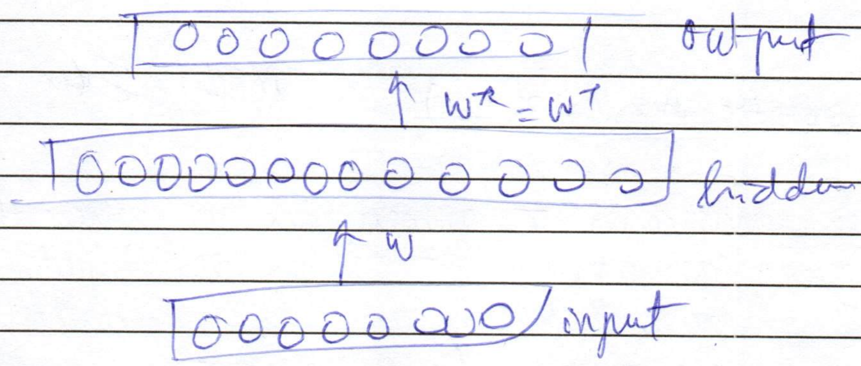
$\therefore \boxed{(W^T - W)\cancel{x} - \lambda I = 0}$

↳ Here we get the equation of PCA

Hence, the subspace is equivalent to PCA.

Overcomplete Autoencoder:—



Here, the hidden layer is not able to identify which features are important because there are more than enough ~~fea~~ neurons in the hidden layer to accomodate all.

Thus, each neuron will just copy the input data ~~and~~, there will be no change as weights ~~as~~ will not be trained.

$\Rightarrow h = x_i$

$o = Wx + b = x_i$

$\therefore$ It just acts as the identity function, output is same as i/p.

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Q3) a) $\partial L / \partial h_{t-k}$ for RNN.

Equation of RNN:–

$$a_t = W_{aa} \cdot a_{t-1} + W_{ax} \cdot x_t + b_a$$
$$y = W_{aa} \cdot a_t + b_y$$

for RNN hidden state = a.

we want to find $\partial L / \partial a_{t-k}$.

$\cancel{\frac{\partial L}{\partial W_{aa}}}$ $\frac{\partial L}{\partial a}$: $L \xrightarrow{\text{depends on}}$ output $\to a_t \to W_{aa}$
$\to a_{t-1} \to W_{aa}$
$\to a_{t-2} \to W_{aa}$
and so on.

∴ we can write:–

$$\frac{\partial L}{\partial a_t} = \cancel{\frac{\partial L}{\partial W_{aa}\, \partial a_n} \times \frac{\partial a_n}{\partial}} \frac{\partial L}{\partial o} \times \frac{\partial o}{\partial a_t} \times \frac{\partial a_t}{\partial W_{aa}} + \frac{\partial L}{\partial o} \times \frac{\partial o}{\partial a_t} \times \frac{\partial a_n}{\partial a_{t-1}} \times \frac{\partial a_{t-1}}{\partial W_{aa}} + \frac{\partial L}{\partial o} \times \frac{\partial o}{\partial a_t} \cdots \times \frac{\partial a_{t-k}}{\partial W_{aa}}$$

$$\therefore \cancel{\frac{\partial L}{\partial W_{aa}} = \partial}$$

$$\therefore \frac{\partial L}{\partial a_t} = \sum_{i=1}^{t} \frac{\partial L}{\partial o} \times \frac{\partial o}{\partial a_i} \times \frac{\partial a_i}{\partial W_{aa}}$$

this is for t = 1 to t
but if we want 1 to ~~t-k~~ k to t

$$\boxed{\frac{\partial L}{\partial a_{t-k}} = \sum_{i=k}^{t} \frac{\partial L}{\partial o} \times \frac{\partial o}{\partial a_i} \times \frac{\partial a_i}{\partial W_{aa}}}$$

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Thus, the equation is derived.

But, we have a vanishing gradient problem.

take for a long equation:-

$$\frac{\partial L}{\partial a_t} = \frac{\partial L}{\partial o} \times \frac{\partial o}{\partial a_t} \times \frac{\partial a_t}{\partial W_{aa}} + \frac{\partial L}{\partial o} \times \frac{\partial o}{\partial a_t} \times \frac{\partial a_t}{\partial a_{t-1}} \times \frac{\partial a_{t-1}}{\partial W_{aa}} + \dots$$

$$\frac{\partial L}{\partial o} \times \frac{\partial o}{\partial a_t} \times \frac{\partial a_{t-1}}{\partial a_{t-2}} \times \frac{\partial a_{t-2}}{\partial a_{t-3}} \times \frac{\partial a_{t-3}}{\dots} \dots$$

continues

here, each $\frac{\partial a_{n-x}}{\partial a_{n-x-1}}$ is in the range 0 to 1, meaning they are very small numbers. Thus, when we multiply many small numbers, the gradient vanishes, becoming close to 0.

Thus, the words at the starting of the sentence ~~have been~~ are given no importance in the output at present.

This problem is solved by LSTM.

LSTM keeps 2 memories, one for long term context and one for short term context.

It manages these memories through the means of gates.

There are 3 gates in LSTM:

1) forget gate
2) input gate
3) Output gate.

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Working of LSTM :-

Cell state $= C_t = f_t \times C_{t-1} + i_t \times \tilde{C}_t$

Candidate Value: $\tilde{C}_t = \tanh(W_{ch} \cdot h_{t-1} + W_{cx} \cdot x_t + b_c)$

h_t = hidden state = $o_t \times \tanh(C_t)$

Forget gate: $\sigma(W_{fh} \cdot h_{t-1} + W_{fx} \cdot x_T + b_f)$

Input gate: $\sigma(W_{ih} \cdot h_{t-1} + W_{ix} \cdot x_T + b_i)$

LSTM prevents the vanishing gradient problem by involving f_t, i_t, which are like filters for information (gates). They perform element-wise multiplication and are able to handle the 2 memories efficiently. The long term memory is separate, and the activations are no longer repeatedly multiplied to cause a vanishing gradient problem. The short term memory is handled differently and the most recent context has more weightage in that.

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Q3) b] GRU (Gated Recurrent Unit) ~~has~~ is a modification of RNN so that the ~~~~ long term context ~~is~~ is maintained and the model can perform well on ~~a~~ longer sentences.

- Unlike the LSTM, GRU holds both the long term memory and the short term memory in the same place.
→ It has only 2 gates - update gate and reset gate.

equations of GRU

We have ~~3~~ equations
1 → Candidate value to ~~the update~~
2 → hidden state which takes into consideration both the previous hidden state and the ~~candidate value~~ input.
3 → update gate - to update information
4 → reset gate → to forget information.

∴ Candidate value:-

$$C_t = \tanh(W_{ch} \cdot h_{t-1} + W_{cx} \cdot x_t + b_c)$$

update gate: $\sigma(W_{uh} \cdot h_{t-1} + W_{ux} \cdot x_t + b_u)$

reset gate = $\sigma(W_{rh} \cdot h_{t-1} + W_{rx} \cdot x_t + b_r)$

(we use sigmoid for update gate and reset gate as the values will lie between 0 and 1, 1 means we want to remember that information and 0 means we want to forget it)
tanh is used for Candidate Value to give result ~~~~ between -1 and 1.

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now the hidden state will hold both long term and short term memory. We want to update info from $\tilde{C}_t$, and keep whatever info isn't updated from previous hidden state.

∴ We can make the eqn:—

$$\boxed{h_t = u_t \times \tilde{C}_t + (1-u_t) \times h_{t-1}}$$

Mathematically, we can see that GRUs have fewer equations, fewer gates so fewer parameters than LSTM.

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	Q1]b]	Overfitting occurs when model capacity is too high. This is because, the model can easily take all the input data, ~~th~~ predict it without any difficulty, which leads to the model only being able to perform on the input data. ~~stop~~ There is high bias, leading to overfitting ∴ the model cannot generalize.

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